- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. Unless otherwise instructed, when a numerical value is needed, use $g = 9.8$.

11 In this question, the unit vector **i** is horizontal and the unit vector **j** is vertically upwards.

A particle of mass 0.8 kg moves under the action of its weight and two forces given by $(k\mathbf{i} + 5\mathbf{j})\,\mathrm{N}$ and $(4\mathbf{i} + 3\mathbf{j})\,\mathrm{N}$. The acceleration of the particle is vertically upwards.

(a) Write down the value of k. **[1]**

Initially the velocity of the particle is $(4\mathbf{i} + 7\mathbf{j})\,\mathrm{m\,s^{-1}}$.

(b) Find the velocity of the particle 10 seconds later. **[4]**